

CASH BURNS

BY ALVAREZ AND LIPPI

Jason Roderick Donaldson

FACT

Current cash holdings determine means of payment

People use cards more when cash holdings are low

QUESTION

Why do cash and credit coexist as means of payment?

How is cash-credit choice determined by level of cash holdings?

ANSWER

Holding cash is costly

Want to minimize cash holdings

Withdraw a small amount of cash when convenient

Run out of cash sometimes, then use credit

KEY ASSUMPTIONS

KEY ASSUMPTIONS

1. Holding cash is costly (R)
2. Using credit is costly (γ)
3. Voluntary cash are withdrawals costly (b)
4. There are some free cash withdrawals

SUPER SIMPLE VERSION OF THE MODEL

SUPER SIMPLE VERSION OF THE MODEL

Continuous time, finite horizon $t \in [0, T]$

Consumer consumes flow $e dt$

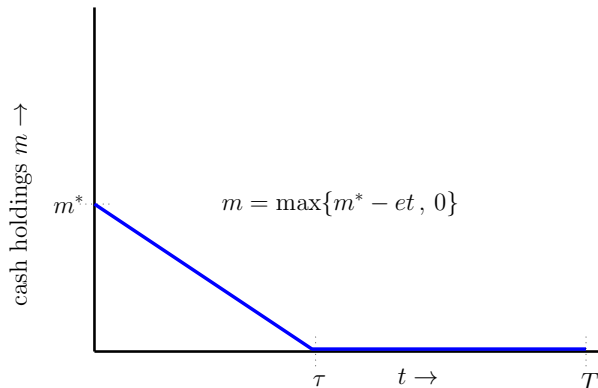
Withdraws m^* in cash at $t = 0$ (no other withdrawals)

Holding cash m costs flow $Rm dt$

Using credit costs $\gamma e dt$

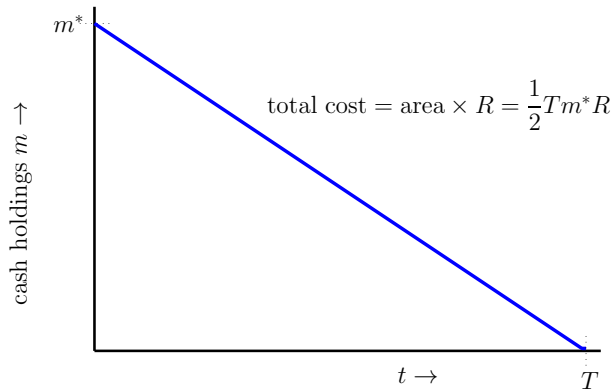
Using is cash free, use credit when run out of cash

CASH HOLDINGS OVER TIME



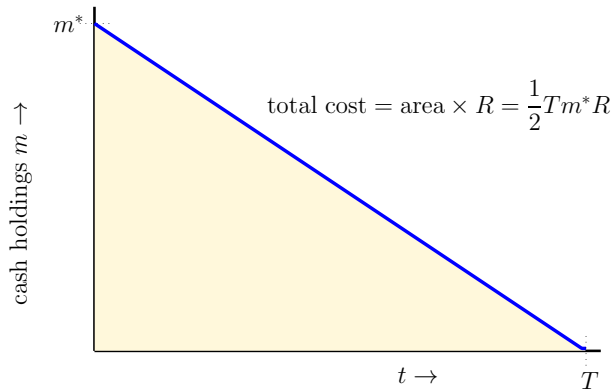
What is the optimal withdrawal size m^* ?

ALL CASH PAYMENTS



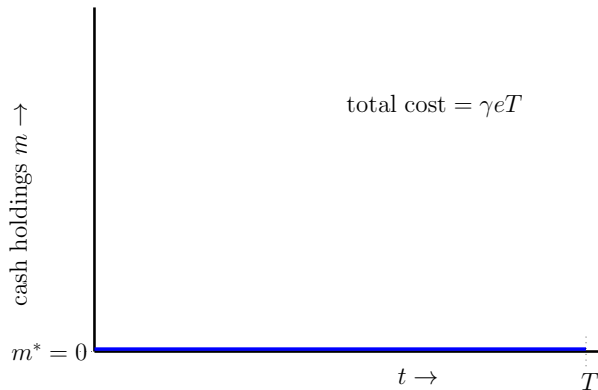
Set $m^* = T/e$, avoid cost of γ using credit

ALL CASH PAYMENTS



Set $m^* = T/e$, avoid cost γ of using credit

ALL CREDIT PAYMENTS



Set m^* zero, avoid cost R of holding cash

CASH OR CREDIT

How does this setup explain coexistence of cash and credit?

Possible conjecture: no coexistence of cash and credit

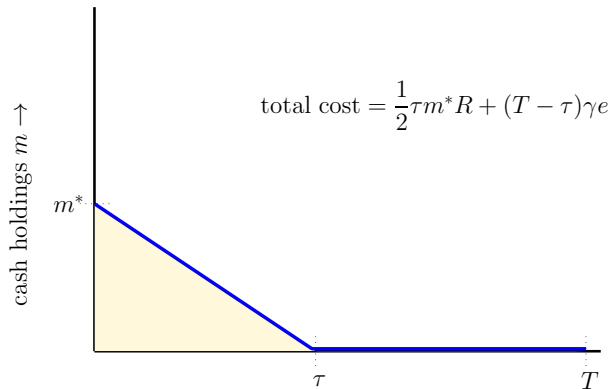
If using credit is expensive (γ large) use cash always

If holding cash is expensive (R large) use credit always

Why is this wrong?

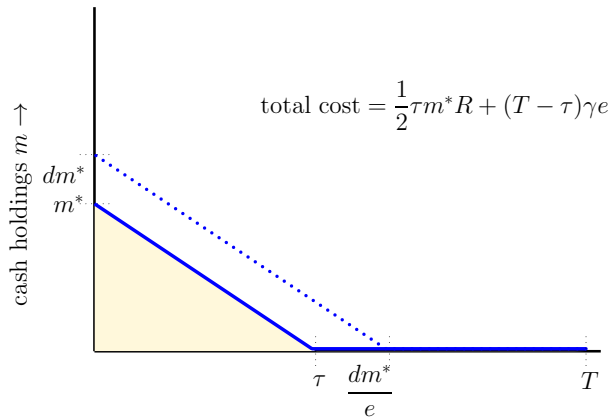
Cash costs convex, credit costs linear $\implies$ mix at optimum

TOTAL COSTS



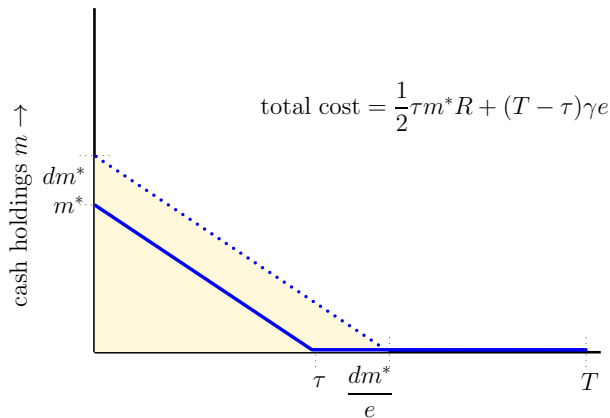
Cash cost proportional to area, credit cost to length

TOTAL COSTS



Cash cost proportional to area, credit cost to length

TOTAL COSTS



Cash cost area increases $\frac{(dm^*)^2}{2e}$, credit cost line shortens $\frac{dm^*}{e}$

SIZE OF WITHDRAWAL m^*

What is the optimal withdrawal?

m^* minimizes

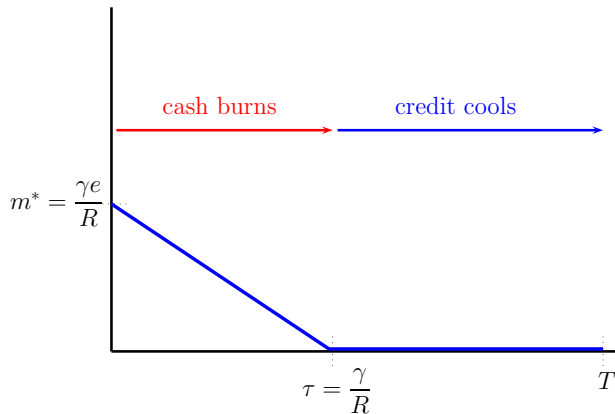
$$\text{total cost} = \frac{1}{2}\tau m^* R + (T - \tau)\gamma e$$

Since $m = \max\{m^* - et, 0\}$, $\tau = m^*/e$ and

$$\text{total cost} = \frac{R(m^*)^2}{2e} - \gamma m^* + T\gamma e$$

So $m^* = \gamma e/R$ and $\tau = \gamma/R$

CASH AND CREDIT



Use cash proportion $\frac{\tau}{T} = \frac{\gamma}{RT}$ of the time

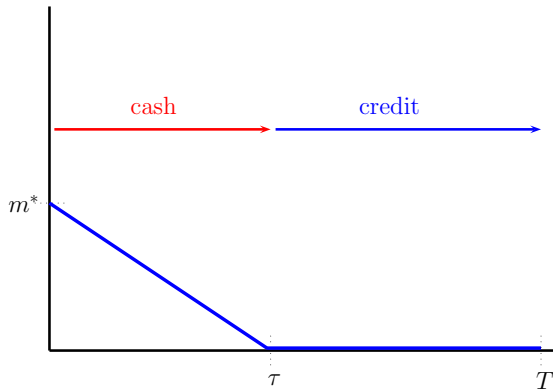
CREDIT IN THE LONG RUN

Because cash burns, use it only for a short time

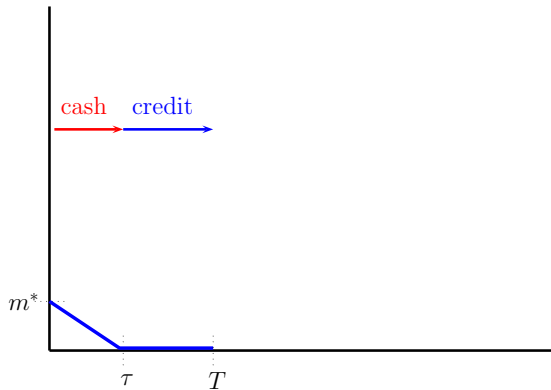
$$\text{prop. time using cash} = \frac{\gamma}{RT} \rightarrow 0$$

as $T \rightarrow \infty$

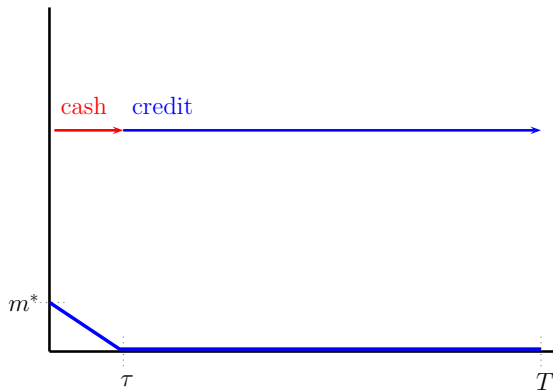
ZOOM IN



ZOOM OUT



LONG HORIZON



Cash becomes negligible for long horizon

ADDING FREE WITHDRAWALS

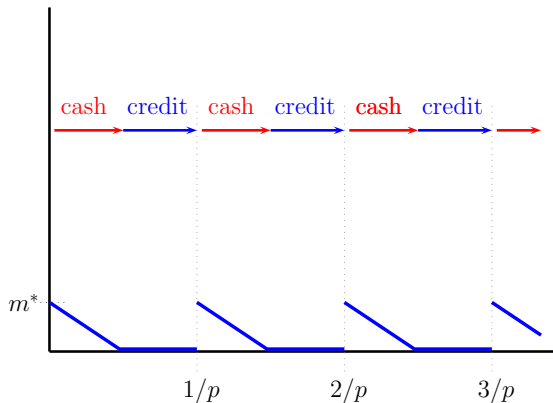
Add free withdrawals at frequency p (Alvarez–Lippi (2009))

You act has if the horizon is $T = 1/p$

I.e. if $t \equiv t' \pmod{1/p}$,

$$m\left(t \mid \text{free withdrawals freq. } 1/p\right) = m\left(t' \mid \text{horizon } 1/p\right)$$

PERIODIC FREE WITHDRAWALS



Switch from cash to credit each period—cash relevant again

COEXISTENCE OF CASH AND CREDIT

Use cash proportion $\tau p = \frac{\gamma}{Rp}$ of the time

Does not depend on horizon

MORE REALISM: RANDOM FREE WITHDRAWALS

With periodic free withdrawals, perfect inventory management

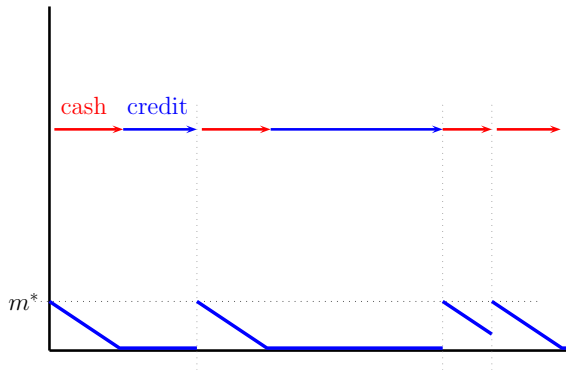
Withdraw only when balance is zero

(Paper lists other “shortcomings” of deterministic model)

Poisson distributed free withdrawals add realism

Top up to m^* at free withdrawal, don't wait to run out

RANDOM FREE WITHDRAWALS



Random free withdrawals: withdraw even if balance is not zero

OPTIONAL COSTLY WITHDRAWALS

Authors also consider costly withdrawal case

Pay fixed cost b to withdraw cash

If no free withdrawals, implies always cash or always credit

If always cash, replenish cash holdings when $m = 0$

Usual (s, S) model

To me: free withdrawals model more interesting/different

SUMMARY SUMMARY

Coexistence of cash and credit in inventory model

Key feature is possibility of free withdrawals

Sequential choice of cash vs. credit (as authors emphasize)

FURTHER COMMENTS

THE ISLAND OF STONE MONEY

WHY IS CREDIT A MEANS OF PAYMENT?

Why did credit evolve as means of payment?

Because the transactions cost of “cash” was high

At least on the Island of Yap (Friedman, *Money Mischief*)

Results here depend on credit being costly, $\gamma > 0$

“time cost of using credit for small value transactions”?

NEGATIVE INTEREST RATES

NEGATIVE INTEREST RATES

Is the cost of holding cash positive?

Negative interest rates suggest yes

But suggest it is a storage/holding cost, not an opportunity cost

POUND NOTES VS. DOLLAR BILLS

POUND NOTES VS. DOLLAR BILLS

Pound notes large and cumbersome compared to dollar bills

Holding cost of cash higher in UK than US

I held cash when I lived in the UK, don't know I live in the US

Why?

UK retailers impose costs of using card

Cash-credit choice determined by other side of market (retailers)

In partial eq., could justify γ

“CAN I GET A CASH BACK?”

“CAN I GET A CASH BACK?”

What happens if can withdraw cash when making card purchase?

Exactly as-is:

Withdraw infinitely frequently but not continuously

I.e. on a dense set of measure zero

100% transactions with cash, but infinitesimal cash holdings

Suppose small costs for cash backs and “free” withdrawals

Back to costly withdrawals—no credit (except to get cash)

CORPORATE CASH

CORPORATE CASH

Why are firms holding so much cash? Here, m^* is high if

Opportunity cost R low = few investment opportunities

Cost of withdrawals high = debt issues costly

Few free withdrawals = low earnings

Suggests countercyclical cash holdings

Apparently so, not precautionary reasons (Stone-Gup (2012))

CONCLUSION

CONCLUSION

Great paper—a pleasure to read

Novel, tractable approach to studying cash and credit together